

eispice: When SPICE Says Hi To Python

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With Python being the EFY DVD theme of the month, we went about exploring software that would make this holy coming together a reality for designers. We landed at eispice, a clone of Berkeley SPICE 3 simulation engine. eispice is optimised for high-speed digital design and has a front-end backed, based and powered by Python.

The design philosophy of Python lends itself to be used for simulators. By design philosophy, I mean the core philosophy that was handily jotted down in the document 'The Zen of Python,' which includes gems such as 'Beautiful is better than ugly,' 'Simple is better than complicated' and others like 'Readability counts.' Such an array of underlying principles is the reason why Python is highly extensible.

SPICE and behind the SPICE

eispice is one of those products that change the landscape of their intended use by virtue of being so good. eispice was initially developed and targeted towards PCB-level signal integrity simulation, simulating Input/output Buffer Information Specification (IBIS) model-defined devices, transmission lines and passive termination. Over the years though, scope of the tool has been expanded to include general-purpose simulation features as well.

Other than being the only open source simulator that provides native IBIS model support, the most unique feature of eispice is its Python based front-end. This is

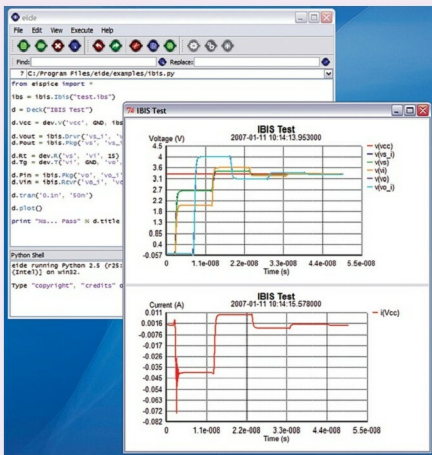


Fig. 1: Simulating IBIS using eispice

achieved by wrapping the simulator into a Python module that makes it possible to use Python language to control simulations and process results. If you are not strong with Python, but do work on Berkeley SPICE, you can use Python shell instead of nutmeg, and Python scripts can be run like batch-mode SPICE.

Features

eispice has a multitude of features, both regular and unique, so as to help you with a variety of tools. The

Quick look at the features

- ▶ Transient and operating-point analysis
- ▶ Independent and non-linear current and voltage source models
- ▶ R, L and C device models
- ▶ Lossless transmission line model and diode model
- ▶ Sub-circuit models available

sub-circuit support allows probing at a deeper level, right to where the sub-assemblies are. eispice also supports a simple Tk based plot-